

PROJECT PLAN & SCHEDULE SPECIFICATION

Project Plan, Work Breakdown & Scheduling Baseline

OneLogin – Single Sign-On Rollout Across a University's Applications

Comprehensive engineering management deliverable detailing the Work Breakdown Structure (WBS), Activity-on-Arrow (AOA) critical path analysis, milestone checkpoints, weekly wave Gantt scheduling, and resource capacity balancing designed to satisfy the strict 16-week pre-semester deadline.

Document Control

Document Title	Project Plan, Work Breakdown Structure & Scheduling Baseline
Document Version	Version 1.0 (Approved Baseline)
Target Schedule Window	16 Weeks (80 Working Days on a 5-day week calendar)
Available Staff	2 Dedicated Integration Engineers (Full-Time Capacity)
Related Deliverables	SRS Document, UML Design Package, Estimation Sheet, Test Plan & Evidence, Risk Register & Closure Note

Formal Sign-Off & Approvals

Role	Designation & Stakeholder	Signature	Approval Date
Project Sponsor	Vice-Chancellor	_____	02 Oct 2026
Business Owner	IT Head	_____	02 Oct 2026
Project Manager	Lead SEPM Consultant	_____	02 Oct 2026
Technical Lead	Lead Systems Engineer	_____	02 Oct 2026

Scheduling Constraint Note: In strict compliance with case parameters, the baseline path commits to **Option A (9 applications: 4 Cloud + 5 In-House)**. The 3 legacy vendor applications (Option B: requiring 45 extra person-days) are excluded from the initial critical path to prevent guaranteed deadline overrun, while a mandatory 10-day blackout window protects student semester examinations.

1. Work Breakdown Structure (WBS)

The Work Breakdown Structure hierarchically partitions all engineering, federation, testing, and operational deployment activities required to deliver the university's centralized identity service.

WBS Code	Work Package Title	Scope & Deliverable Specification
1.0	OneLogin SSO Implementation	End-to-end identity federation and rollout across university applications.
1.1	Initiation & Requirements	Stakeholder elicitation interviews, legacy ticket analysis, IEEE 830 baseline.
1.1.1	Requirements Elicitation	Interviews with students, staff, helpdesk, and application owners.
1.1.2	SRS Specification Sign-Off	Definition of 11 FRs, 6 measurable NFRs, MoSCoW prioritization, and RTM.
1.2	Architecture & System Design	UML structural and behavioral models, interface decoupling, flowcharts.
1.2.1	Architectural Flowcharts	Central gate authentication decision flow, lockout logic, and MFA branching.
1.2.2	UML Modeling Package	Use cases, class diagrams, 3-app sequence flow, and session state machines.
1.3	Infrastructure & Core IdP	Staging host environment, TLS 1.2+ configuration, directory binding.
1.3.1	IdP Server Staging	Active-standby high-availability instances, Redis session cache, TLS certs.
1.3.2	Directory Synchronization	LDAP/ERP attribute mapping, secondary email/SMS contacts, RBAC tables.
1.4	Application Federation Waves	Staged integration of university applications grouped by protocol complexity.
1.4.1	Wave 1: Cloud Apps (4 apps)	SAML 2.0 / OIDC federation (3 days per app = 12 person-days total effort).
1.4.2	Wave 2: In-House Apps (5 apps)	Local login removal, OIDC token callback (8 days/app = 40 person-days).
1.4.3	Wave 3: Legacy Vendor Apps (3)	Reverse-proxy credential adapter (15 days/app = 45 person-days; Gated).
1.5	Testing & Quality Verification	Functional test suite (TC-01 to TC-23), security audits, and load profiling.
1.5.1	BVA & Access Rule Testing	Failed attempts (4, 5, 6), session timeout (29m, 30m, 31m), OTP expiry (15m).
1.5.2	Performance & Recovery Drills	Response profiling (p95 ≤ 2 s), synthetic uptime, RTO (≤30m) & RPO (≤15m) drills.
1.6	Training, Pilot & Cutover	Operational transition, end-user self-service rollout, and cutover buffer.
1.6.1	Helpdesk & SSPR Enablement	Helpdesk unlock console, staff training, self-service password reset pilot.
1.6.2	Production Cutover & Buffer	Staged cohort launch, final cutover, and 5-day post-launch warranty buffer.

2. Activity Table, Critical Path & Float Calculations

The schedule models dependencies across 2 dedicated integration engineers (E₁, E₂) over an 80-working-day timeframe. Forward and backward passes compute early and late schedules, identifying total float and establishing the project's critical path.

ID	Activity Description	Effort	Dur	Res	Pred	ES	EF	LS	LF	Float	Critical?
A1	Requirements & SRS Approval	10 p-d	5 d	E ₁ , E ₂	None	0	5	0	5	0 d	CRITICAL
A2	Architecture & UML Package	10 p-d	5 d	E ₁ , E ₂	A1	5	10	5	10	0 d	CRITICAL
A3	Core IdP & Directory Binding	12 p-d	6 d	E ₁ , E ₂	A2	10	16	10	16	0 d	CRITICAL
A4	Wave 1: Cloud Apps (4 apps)	12 p-d	6 d	E ₁ , E ₂	A3	16	22	16	22	0 d	CRITICAL
A5	Wave 2: In-House Apps (5 apps)	40 p-d	20 d	E ₁ , E ₂	A4	22	42	22	42	0 d	CRITICAL
A6	Integration & Security Testing	16 p-d	8 d	E ₁ , E ₂	A5	42	50	42	50	0 d	CRITICAL
A7	Exam Blackout (Zero Outages)	0 p-d	10 d	None	A6	50	60	50	60	0 d	CRITICAL
A8	Helpdesk Training & SSPR	5 p-d	5 d	E ₁	A7	60	65	65	70	5 d	FLOAT: 5D
A9	Pilot Rollout & UAT	16 p-d	8 d	E ₁ , E ₂	A7	60	68	62	70	2 d	FLOAT: 2D
A10	Production Cutover	10 p-d	5 d	E ₁ , E ₂	A8, A9	68	73	70	75	2 d	FLOAT: 2D
A11	Warranty Buffer & Close	10 p-d	5 d	E ₁ , E ₂	A10	73	78	75	80	2 d	FLOAT: 2D

Mathematical CPM Calculation Summary

- **Forward Pass:** Early Finish ($EF = ES + \text{Duration}$); Early Start equals the maximum Early Finish of all immediate predecessors ($ES = \max\{EF_{pred}\}$).
- **Backward Pass:** Late Start ($LS = LF - \text{Duration}$); Late Finish equals the minimum Late Start of all immediate successors ($LF = \min\{LS_{succ}\}$).
- **Total Float:** Computed via $Float = LS - ES = LF - EF$. Tasks exhibiting zero total float govern the primary critical sequence.
- **Identified Critical Path:** A1 → A2 → A3 → A4 → A5 → A6 → A7 → A9 → A10 → A11.
- **Project Completion:** Total elapsed working duration is **78 working days** (15.6 calendar weeks), fitting securely inside the 80-working-day ceiling with 2 days of unallocated reserve.

3. Activity-on-Node (AON) Network Diagram

The network diagram models the dependency graph across all work packages. Red highlighted nodes represent activities on the Critical Path (Float = 0). Blue highlighted paths reflect parallel tasks with available schedule float.

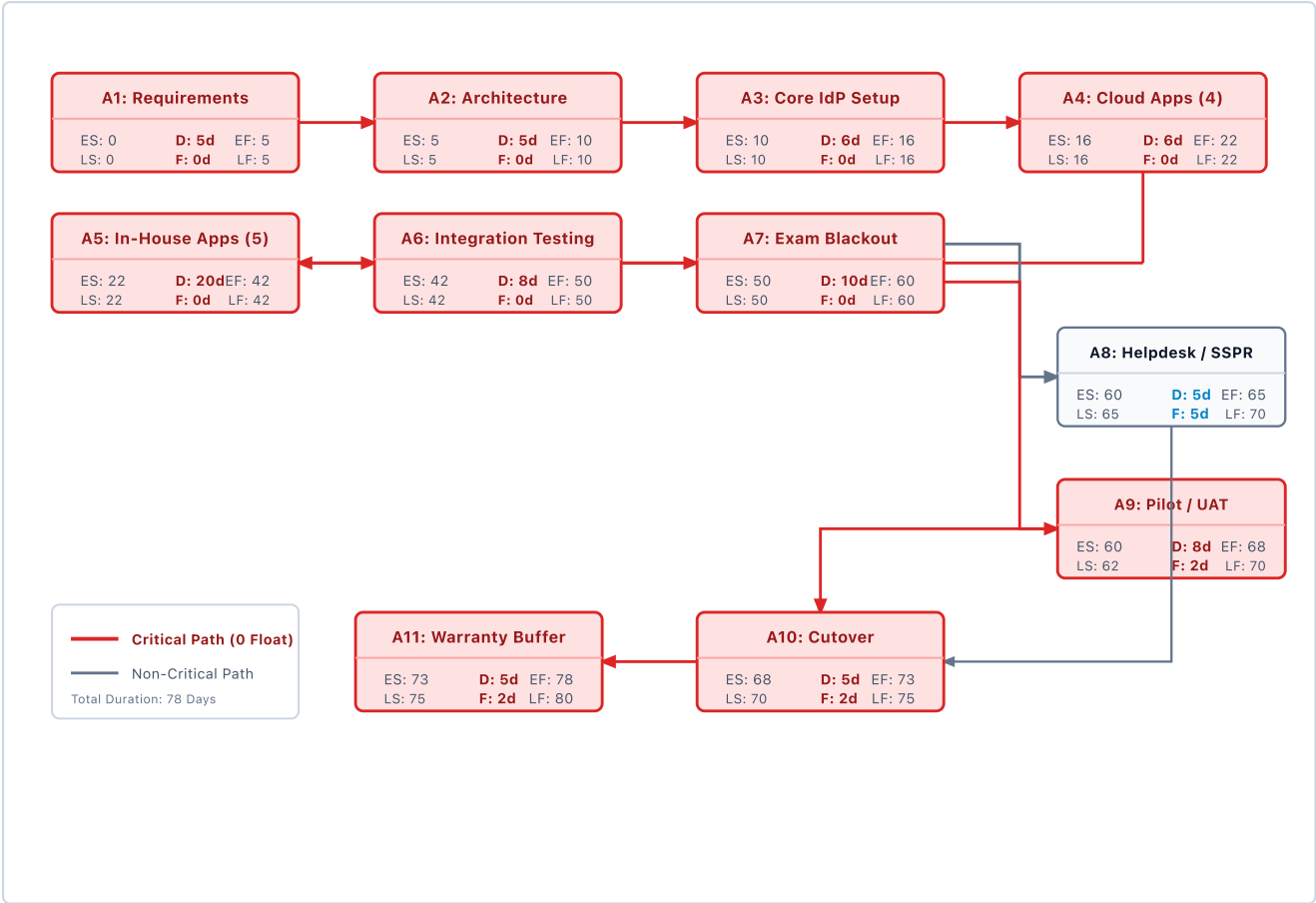


Figure 3.1: Activity-on-Node (AON) Network Diagram with Forward/Backward Pass Data.

4. Gantt Schedule & Milestone Checkpoints

The weekly implementation wave tracks execution across the 16 project weeks. Zero-duration checkpoints (milestones) define mandatory verification gates before downstream deployment phases initiate.

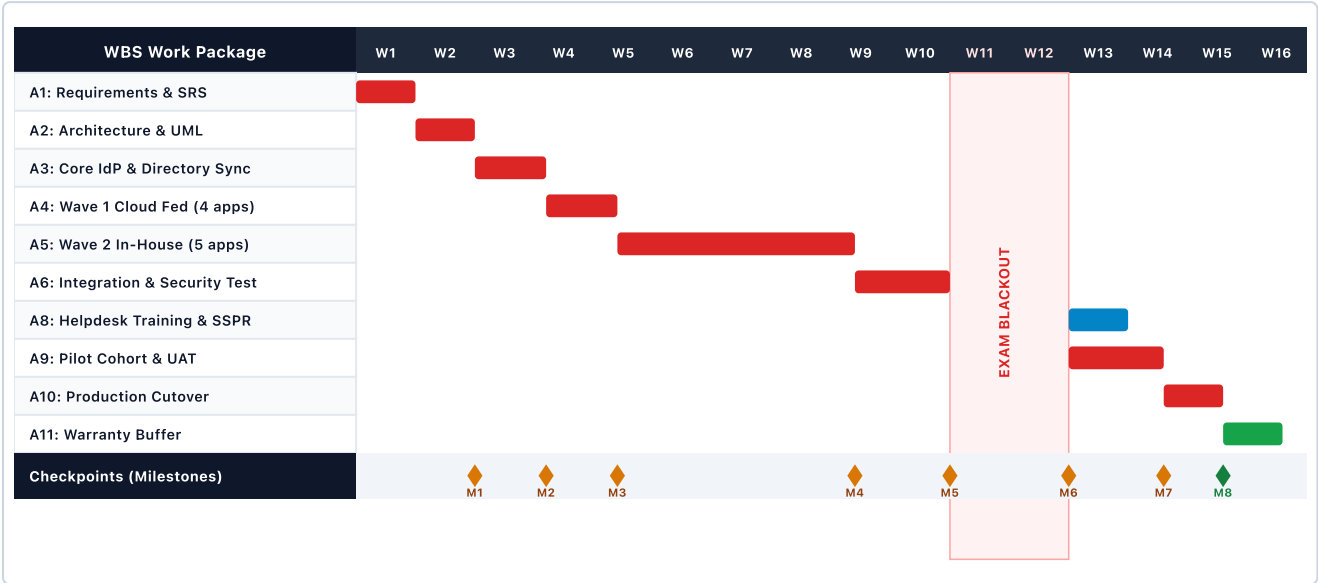


Figure 4.1: 16-Week Schedule Indicating Activity Waves, Critical Red Line, and Exam Blackout.

ID	Milestone Description	Target Timeline	Verification Deliverable & Quality Gate
M1	Baseline Approvals	Day 10 (End W02)	SRS Document and UML Design Package formally approved.
M2	Directory Sync Complete	Day 16 (Mid W04)	LDAP/ERP schema binding and secondary contact extraction verified.
M3	Cloud Federation Wave	Day 22 (End W04)	4 Cloud applications authenticated via SAML 2.0 / OIDC in staging.
M4	In-House Integrations	Day 42 (Mid W09)	5 In-house apps live in staging; local login forms disabled.
M5	Integration Testing Gate	Day 50 (End W10)	Zero critical defects; test suite (TC-01 to TC-23) passed. Pre-exam freeze.
M6	Exam Blackout Exit	Day 60 (End W12)	Zero production disruptions during student examination period.
M7	Pilot UAT Sign-Off	Day 68 (Mid W14)	SSPR self-service success verified by student/staff pilot cohort.
M8	Semester Go-Live	Day 73 (Mid W15)	Production cutover complete; 9 applications under centralized SSO.

5. Resource Allocation & Scope Justification

The team structure balances specialized engineering responsibilities between the 2 dedicated resources to prevent project delivery bottlenecks.

Resource Role	Key Responsibilities & Assigned Modules	Allocation	Workload
Engineer 1 (Lead Protocols & Infrastructure)	Core IdP server staging, directory synchronization, TLS certificate binding, SAML/OIDC cloud connectors, failover drills, security scans, helpdesk unlock console tooling.	Weeks 1–16 (Dedicated)	78 Days (100% Load)
Engineer 2 (Full-Stack & Application Integrator)	In-house application refactoring, legacy form disabling, session callback handling, UI launcher implementation, SSPR workflow, automated test scripting (TC-01 to TC-23).	Weeks 1–16 (Dedicated)	78 Days (100% Load)
Application Owners (University IT Staff)	Providing staging access, assisting with codebase refactoring, reviewing application-level access matrix permissions.	Weeks 4–9 (Part-Time)	Advisory & Review Only
Helpdesk Lead (Operations Representative)	Helpdesk unlock runbook review, participating in SSPR usability drills, validating account lockout release workflows.	Weeks 13–15 (Part-Time)	Operational Acceptance

Strategic Scope Trade-Off: Option A vs. Option B

The problem statement dictates rolling out single sign-on across the university's applications prior to the next semester, exactly 16 weeks (80 working days) away, using 2 engineers.

- Option A (9 Applications - 4 Cloud + 5 In-House):**
Integration effort equals $(4 \times 3) + (5 \times 8) = 52$ person-days. Combined with requirements (10 p-d), design (10 p-d), core setup (12 p-d), testing (16 p-d), training/pilot (26 p-d), cutover (10 p-d), and post-launch buffer (10 p-d), total project work effort is **136 person-days**. With 2 engineers, this requires 68 elapsed working days. Adding the 10-day exam blackout yields **78 working days** (15.6 weeks). This completes within the 80-day deadline with a 2-day contingency reserve.
- Option B (12 Applications - Including 3 Legacy Vendor Apps):**
Adding the 3 old vendor products with no native SSO adds $3 \times 15 = 45$ person-days of custom adapter development, expanding testing by 8 person-days. Total work effort reaches **189 person-days**, requiring 94.5 elapsed working days. Adding the 10-day exam blackout pushes total elapsed duration to **104.5 working days (20.9 calendar weeks)**.

Executive Project Conclusion: Attempting Option B with 2 engineers causes a fatal **4.9-week project overrun** beyond the semester deadline. Option A is therefore the committed engineering baseline. The 3 vendor applications are deferred to a Phase 2 extension post-semester.